

GIO: Generating Efficient Matrix and Frame Readers for Custom Data Formats by Example

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Data Scientists deal with a wide variety of file data formats and data representations. Probably the most difficult to handle are custom data formats that liberally define their own particular flat or nested structure with multiple custom delimiters, multi-line records, or undocumented semantics of attribute sequences, co-appearances, and repetitions. As a prerequisite for exploratory ML model training, data scientists need to map these data representations into regular frames or matrices. Unfortunately, existing tools and frameworks provide only limited support for aiding this process, which causes redundant manual efforts and unnecessary data quality issues. In this paper, we initiate work on automatic matrix and frame reader generation by example. A user provides a sample of raw text data and its mapped matrix or frame representation. Our GIO framework then first identifies the mapping rules from raw to structured data, and subsequently generates source code of an efficient, multi-threaded reader for reading full raw datasets of this format. In order to facilitate manual improvements, both the mapping rules, and generated reader can be modified as needed. Our experiments show that GIO is able to correctly identify the mapping rules for basic text formats like CSV, LibSVM, MatrixMarket; custom text formats from publishing, automotive, and health care; as well as various nested formats such as JSON and XML. Additionally, the automatically generated readers yield competitive performance compared to hand-coded readers and tuned libraries like RapidJSON.

CCS Concepts: • **Information systems → Database management system engines**; **Data scans**; **Record and block layout**; • **Theory of computation → Database query processing and optimization (theory)**.

Additional Key Words and Phrases: Raw Data; Custom Data Format; Efficient Readers; Data Loading

ACM Reference Format:
Saeed Fathollahzadeh and Matthias Boehm. 2023. GIO: Generating Efficient Matrix and Frame Readers for Custom Data Formats by Example. *Proc. ACM Manag. Data* 1, 2, Article 120 (June 2023), 26 pages. https://doi.org/10.1145/3589265

1 INTRODUCTION

The typical data science lifecycle is exploratory, where data scientists formulate hypotheses, integrate and clean the necessary data in order to build and evaluate predictive models [31, 36]. Data sourcing and integration often deals with files in open data formats, stored in cloud object storage or distributed file systems [79]. Common formats include text formats such as CSV, fixed-width text, JSON, and XML; as well as binary formats such as Parquet, HDF5, and Protobuf. Apart from these domain-agnostic, syntactic data formats, there are also a number of domain-specific, semantic

*This work was partially done at Graz University of Technology, Austria.

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2836-6573/2023/6-ART120 $15.00
https://doi.org/10.1145/3589265